

Mini ERP + CRM Operations Portal

Full-Stack Technical Solution & Architecture Report

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1. Executive Summary & Business Context

This document presents the complete engineering solution for the Fundsroom Full Stack Developer Case Study: a production-grade Mini ERP + CRM Operations Portal developed for a wholesale distribution enterprise. The enterprise model requires coordinating client acquisition, catalog inventory with alert thresholds, tamper-evident stock movement logs, and sales delivery challans.

Key engineering achievements in this submission include:

- **Role-Driven Architecture:** Distinct, decoupled workspaces and login experiences for Executive Administrators versus Staff members (Sales, Warehouse, Accounts).
- **Concurrency Safety:** Atomic stock deduction with PostgreSQL row-level locks (SELECT ... FOR UPDATE), guaranteeing zero negative inventory and no race conditions.
- **Historical Snapshot Integrity:** Preserving product name, SKU, and unit price in challan_items to maintain historical accuracy regardless of future catalog changes.
- **Bonus Implementations:** Full containerization via Docker Compose, automated CI/CD via GitHub Actions, and browser-native PDF delivery note export.
- **Immediate Joining Commitment:** I confirm my 100% availability for immediate joining in Pune, Maharashtra upon selection.

2. Tools & Technologies Used & How They Work

The project leverages modern industry tools chosen specifically for speed, reliability, type safety, and scalability:

Tool / Technology	Layer	How It Works & Business Value
React 18 & TypeScript	Frontend Tier	Provides a robust Single Page Application (SPA) architecture with strict type safety, modular component lifecycle management, and instant sub-second bundling with Vite.
Node.js 24 & Express.js	Backend REST API	Asynchronous, event-driven RESTful micro-service with express-validator validation pipelines, role authorization middleware, and centralized error handling.
PostgreSQL (Neon Cloud AWS)	Database Tier	Serverless PostgreSQL engine providing ACID transactional integrity, multi-client connection pooling, and row-level locking

primitives.

JSON Web Tokens (JWT) & Bcrypt	Security & Auth	Stateless authentication using cryptographically signed JWT tokens with 24-hour expiration, paired with salted bcrypt hashing (10–12 salt rounds) for password storage.
Docker & Docker Compose	Containerization (Bonus)	Multi-stage Dockerfiles for backend (Node 22-alpine) and frontend (Nginx alpine) orchestrated via docker-compose.yml with local PostgreSQL networking.
GitHub Actions	CI / CD Pipeline (Bonus)	Automated continuous integration pipeline (.github/workflows/ci.yml) validating TypeScript compilation, linting, and production builds on push/PR.
Postman Collection	API Testing (Bonus)	Structured collection (postman/Fundsroom-ERP.postman_collection.json) covering 100% of REST endpoints across auth, CRM, products, and challans.
Render Cloud Hosting	Cloud Deployment	Configured for zero-downtime deployment on Render.com with single-service production serving (Express serving client/dist static build).

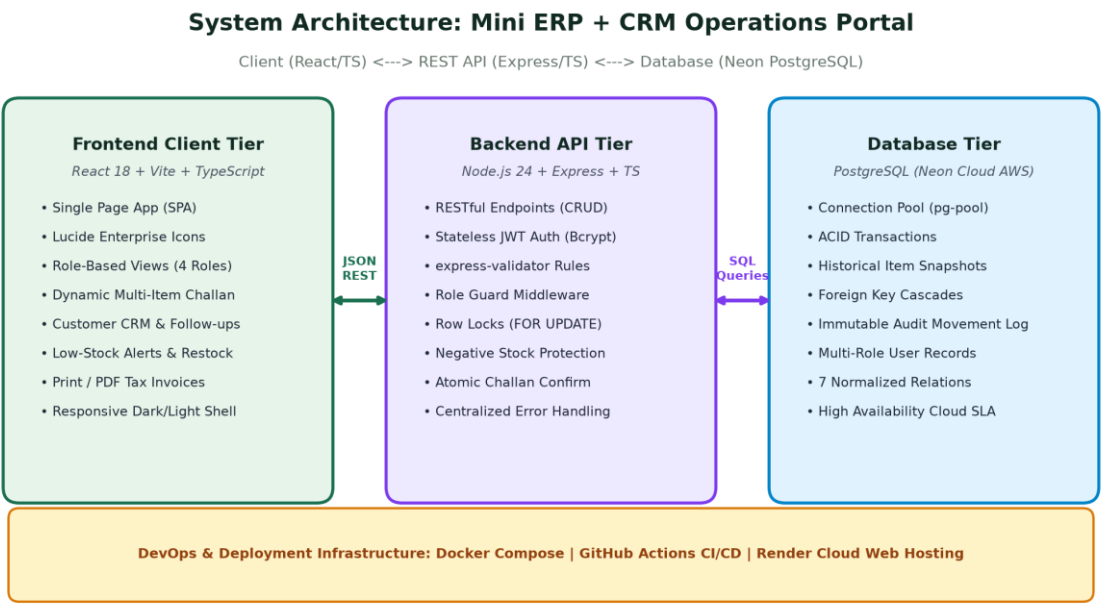


Figure 1: Full-Stack System Architecture (React Frontend, Express API, Neon PostgreSQL DB, and Docker/CI Infrastructure)

3. Database Schema & Data Integrity Design

The database architecture is designed with 7 normalized tables, enforcing relational foreign key constraints, automatic timestamps, and check constraints to prevent invalid operational states:

- **users:** Stores employee credentials, salted password hashes, and strictly enumerated roles (Admin, Sales, Warehouse, Accounts).
- **customers:** Complete CRM records including business name, mobile, email, customer type (Retail/Wholesale/Distributor), status (Lead/Active/Inactive), follow-up date, and optional GSTIN.
- **customer_followups:** Chronological interaction history linked to customers with created_by employee foreign key.
- **products:** Catalog SKUs, categories, unit prices (NUMERIC 12,2), on-hand stock (CHECK >= 0), minimum alert thresholds, and warehouse location.
- **stock_movements:** Immutable audit trail tracking product, movement direction (IN/OUT), quantity changed (> 0), reason reference, and authorized employee.
- **challans:** Outbound delivery orders with unique formatted challan numbers (CH-YYYY-XXXXXX), customer relation, total items, and status (Draft/Confirmed/Cancelled).
- **challan_items:** Historical product snapshot preserving product name, SKU, and unit price at moment of sale.

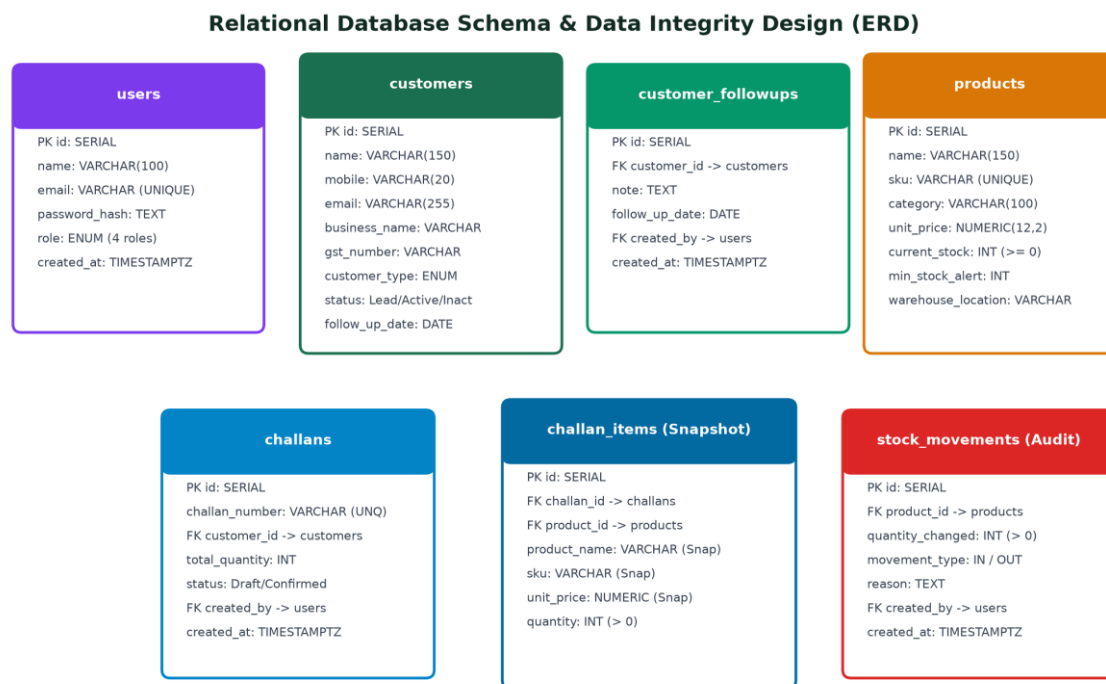


Figure 2: Relational Database Entity Relationship Diagram (ERD) with Constraints & Snapshots

4. Transaction Safety & Concurrency Control (Stock Safety)

In high-volume wholesale distribution, race conditions during order placement can lead to inventory overselling or negative stock balances. OrbitOps solves this at the database engine level:

- When a sales challan is confirmed, Express initiates an atomic transaction using BEGIN.
- PostgreSQL Row-Level Locking: Executes `SELECT current_stock FROM products WHERE id = $1 FOR UPDATE`. This locks the specific product rows against concurrent modifications.
- Zero Negative Stock Check: If `current_stock < requested_quantity`, the transaction immediately executes ROLLBACK and returns HTTP 409 Conflict with an error message.
- Atomic Stock Decrement: Deducts on-hand inventory via `UPDATE products SET current_stock = current_stock - $1 WHERE id = $2`.
- Automatic Audit Log: Concurrently writes an OUT entry in `stock_movements` linked to the confirmed challan number.

- Atomic COMMIT commits all updates together, ensuring no partial or corrupted states.

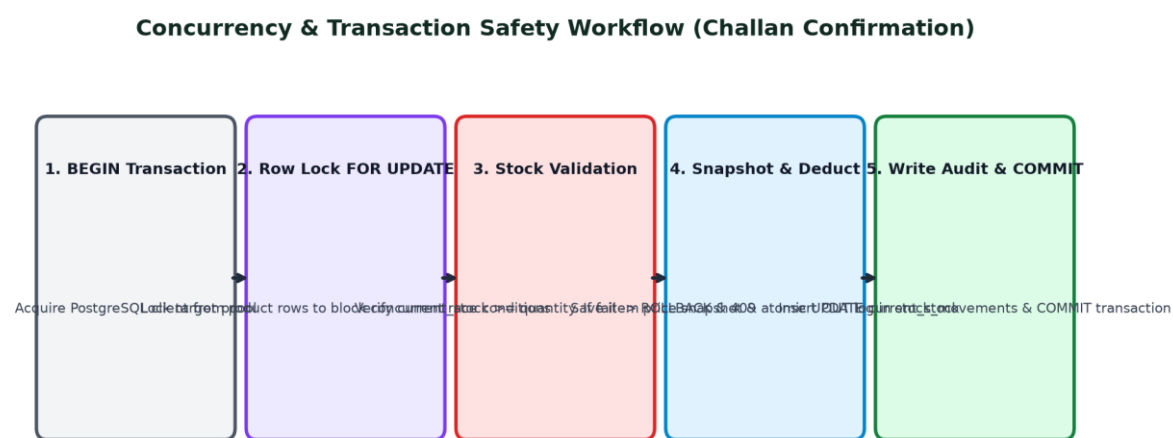


Figure 3: Five-Step Atomic Transaction Workflow with PostgreSQL Row-Level Locking (FOR UPDATE)

5. Role-Based Access Control: Admin vs. Staff Experience

Per the prompt instructions, distinct workspaces and portals were engineered for Executive Administrators versus Department Staff:

A. Login Page Role Separation

The sign-in interface features a Tabbed Entrance separating the Admin Portal from the Staff Workspace. Administrators enter through the purple Executive Portal, while staff select their respective department. One-click demo login buttons and a password visibility toggle (Eye icon) allow instant testing.

B. Executive Admin Portal Experience

When logging in as Admin (admin@fundsroom.local):

- Executive Command Center: Dashboard displaying high-level operational health, inventory asset value, live low-stock warnings, and recent dispatches.
- Exclusive 'Team & Roles' Governance (/admin/users): Administrators have exclusive access to view all registered employees and dynamically reassign or promote roles (Admin, Sales, Warehouse, Accounts) via dropdowns.
- Unrestricted Access: Admin can perform all operations across CRM, Catalog, Restock, Dispatches, and Auditing.

C. Department Staff Workspaces (Sales, Warehouse, Accounts)

When logging in as Staff:

- Sales (sales@fundsroom.local): Dedicated workspace for Customer CRM leads, follow-up scheduler, and multi-product Challan generation. Cannot modify warehouse stock directly.
- Warehouse (warehouse@fundsroom.local): Dedicated logistics workspace with on-hand inventory, low-stock alerts, 1-click Quick Restock, and goods inward logs. Cannot view customer CRM.
- Accounts (accounts@fundsroom.local): Read-only financial auditing workspace with delivery notes, GSTIN verification, and printable tax invoices.

Role-Based Access Control (RBAC) Permission Matrix

Functional Module / Action	Admin	Sales	Warehouse	Accounts
Customer CRM (Add / Edit)	Full Access	Full Access	Read-Only	Read-Only
Customer Follow-Up Loggi	Full Access	Full Access	Forbidden	Read-Only
Products Catalog (Add / E	Full Access	Read-Only	Full Access	Read-Only
Inventory Quick Restock	Full Access	Forbidden	Full Access	Forbidden
Sales Challan (Create Mult	Full Access	Full Access	Forbidden	Read-Only
Confirm Challan & Stock D	Full Access	Full Access	Forbidden	Forbidden
Stock Audit Trail (IN / OUT	Full Access	Read-Only	Full Access	Read-Only
Team Member Role Reassi	Admin Exclusive	Forbidden	Forbidden	Forbidden

Figure 4: Comprehensive Role-Based Access Control (RBAC) Permission Matrix across All 4 Roles

6. Development Environment & Repository Evidence

The development environment in Visual Studio Code and the official GitHub repository setup are shown below:

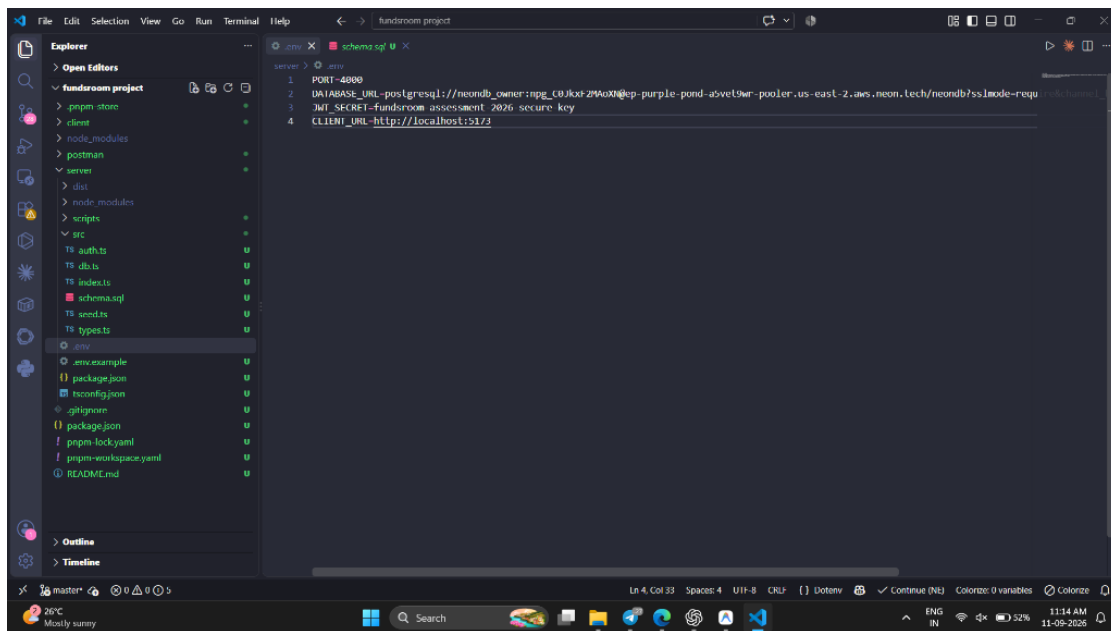


Figure 5: Local VS Code Workspace with Neon PostgreSQL Active Database Configuration (.env) & Monorepo Structure

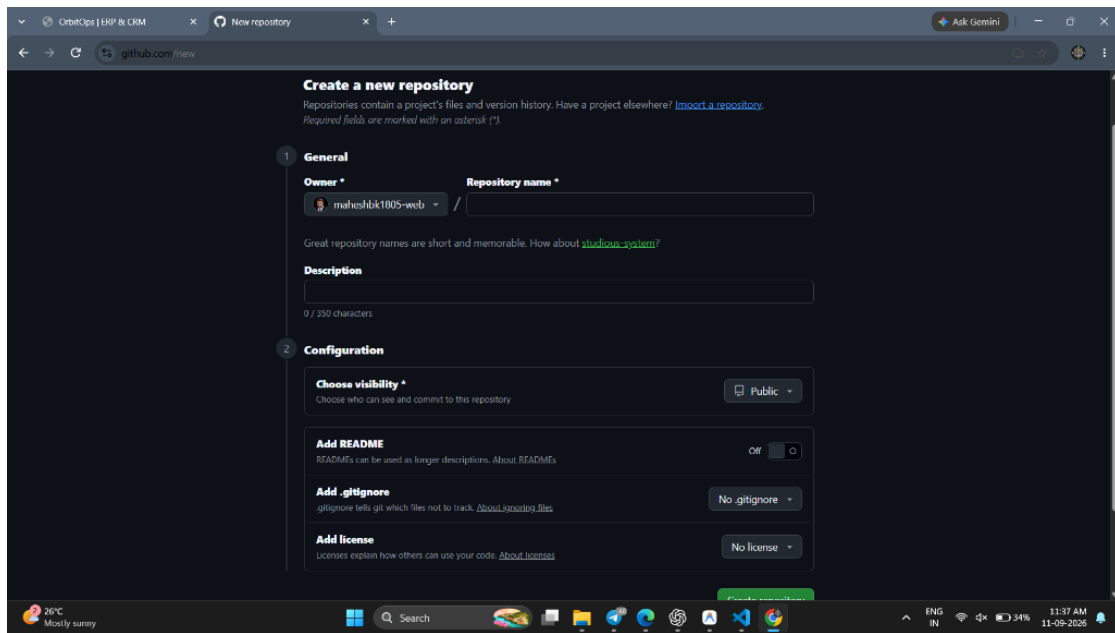


Figure 6: Official GitHub Repository Creation for fundsroom-mini-erp-crm (maheshbk1805-web)

7. Verified Test Login Credentials

All accounts are pre-seeded in the Neon Cloud database with password:

Universal Password: Password@123

Role	Login Email	Password	Workspace Access Scope
Admin	admin@fundsroom.local	Password@123	Executive full access: CRM, Inventory, Orders, Audit logs, and exclusive User Role Governance.
Sales	sales@fundsroom.local	Password@123	Sales domain: Register clients, log follow-ups, draft and confirm multi-item delivery challans.
Warehouse	warehouse@fundsroom.local	Password@123	Logistics domain: Manage catalog products, set reorder alert thresholds, 1-click Quick Restock.
Accounts	accounts@fundsroom.local	Password@123	Auditing domain: Read-only operational visibility across all records and printable delivery invoices.

8. Screen Recording & Google Form Submission Checklist

Per the case study instructions, follow this exact procedure for the submission:

- 1. **Screen Recording:** Start your screen recorder (Windows Game Bar [Win + G], OBS Studio, or Loom).

- **2. Admin Flow:** Demonstrate Admin login, Executive Command Center metrics, and the User Management page.
- **3. CRM Flow:** Go to Customer CRM &arr; Add customer with GSTIN &arr; Open Profile &arr; Log a follow-up note.
- **4. Inventory Flow:** Go to Inventory &arr; Show low-stock alert badge &arr; Click '+ Restock' to inward goods in 1 click.
- **5. Challan Flow:** Go to Sales Challans &arr; Click 'New Challan' &arr; Add multiple products &arr; Create Confirmed &arr; Show printable Tax Invoice.
- **6. Audit Log:** Go to Stock Audit Log &arr; Show the automated OUT movement logged for the confirmed challan.
- **7. Role Switching:** Sign out &arr; Log in as Sales or Accounts to showcase the tailored role workspace.
- **8. Drive Upload:** Upload your video recording to Google Drive. Right-click &arr; Share &arr; Set General Access to 'Anyone with the link can view'.
- **9. Form Submission:** Submit the Google Form before 6:00 PM with your Drive link and this PDF document.

9. Immediate Joining Commitment

I, Mahesh B K, confirm my genuine interest and availability for immediate joining at Fundsroom in Pune, Maharashtra. I am fully comfortable with the Pune work location and look forward to clearing Rounds 2 and 3 and joining the organization without fail.